Cisco Packet Tracer Mini Project Report

Manipal Institute of Technology, Bengaluru

Project title: Design and Simulation of a Secure, Fault-Tolerant Hospital Network using Cisco Packet Tracer

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1 Problem Definition

A mid-sized hospital has five departments (Reception, Doctors/OPD, Laboratory, Pharmacy and a server room) plus a guest Wi-Fi area in the lobby. Staff PCs must reach shared servers (patient records, a staff portal, DNS/DHCP) and the internet, but each department should only reach what it needs, and guests must be kept completely away from patient data. A flat network, with every device in one broadcast domain behind one switch and one gateway, fails on both counts: any user can reach the records server, and a single switch or cable failure stops the whole hospital. This project designs and simulates in Cisco Packet Tracer 9.0.1 a network that separates the departments, controls traffic between them, survives the loss of a core switch or link, and gives staff and guests internet access, and then verifies all of this with ping, web, FTP and SSH tests.

2 Objectives

- Connect all end devices, access switches, core switches and routers in a three-layer (access, core, edge) design.
- Separate the departments into VLANs and assign addresses with a VLSM plan from 172.16.10.0/23.
- Make the core fault tolerant with two multilayer switches, dual uplinks, an LACP EtherChannel, Rapid-PVST and HSRP.
- Route between VLANs and to the internet with OSPF and PAT, and provide DHCP, DNS, web and FTP services.
- Enforce a security policy with ACLs, port security, unused-port shutdown and SSH-only management.
- Verify communication, the access policy and failover between the specified devices with tests.

3 Network Requirements

The devices below were placed in Packet Tracer. Every access switch has one uplink to each core switch, the two cores are joined by a two-link EtherChannel, and the edge router has a routed link to each core.

Table 1: *Devices used*

Device (model)	Qty	Name(s)	Role
Multilayer switch (Catalyst 3560-24PS)	2	CORE1, CORE2	Core layer: inter-VLAN routing, HSRP gateways, ACLs
Switch (Catalyst 2960-24TT)	5	SW-RECEP, SW-DOC, SW-LAB, SW-PHARM, SW-SRV	Access layer, one per department / server room
Router (ISR 2911)	2	EDGE, ISP	EDGE: OSPF, default route, PAT. ISP: simulated provider
Server-PT	4	RECORDS, PORTAL, DNS-DHCP, INET-SRV	Patient records (web, FTP), staff portal, DNS/DHCP, internet web server
PC-PT	9	PC-RECEP1/2, PC-DOC1/2, PC-LAB1/2, PC-PHARM1/2, ADMIN-PC	Department users and the network admin PC
AccessPoint-PT + wireless clients	1 + 2	AP-LOBBY, GUEST-LAPTOP, GUEST-PHONE	Guest Wi-Fi in the lobby

Links: copper straight-through cables to end devices and the edge router, crossover cables between switches, 25 devices and 29 cabled links in total. **Subnets:** 7 VLAN subnets, 2 point-to-point /30 core-to-edge links, a public /30 to the ISP and a simulated internet LAN (Table 3).

Devices and services used: VLANs and 802.1Q trunks, VLSM, LACP EtherChannel, Rapid-PVST, HSRP, inter-VLAN routing on SVIs, OSPF area 0, static default route, PAT (NAT overload), DHCP with relay, DNS, HTTP/HTTPS, FTP, extended named ACLs, port security, SSH version 2.

4 Network Topology

Figure 1 shows the completed topology. The **access layer** connects the users, servers and the lobby access point. The **core layer** has two 3560 switches that route between VLANs. If one core switch or uplink fails, traffic moves to the other one. The **edge** router connects the hospital to the ISP. Dashed lines are crossover links. On each access switch the uplink to CORE2 is blocked by spanning tree (amber), and it takes over only if the CORE1 path fails.

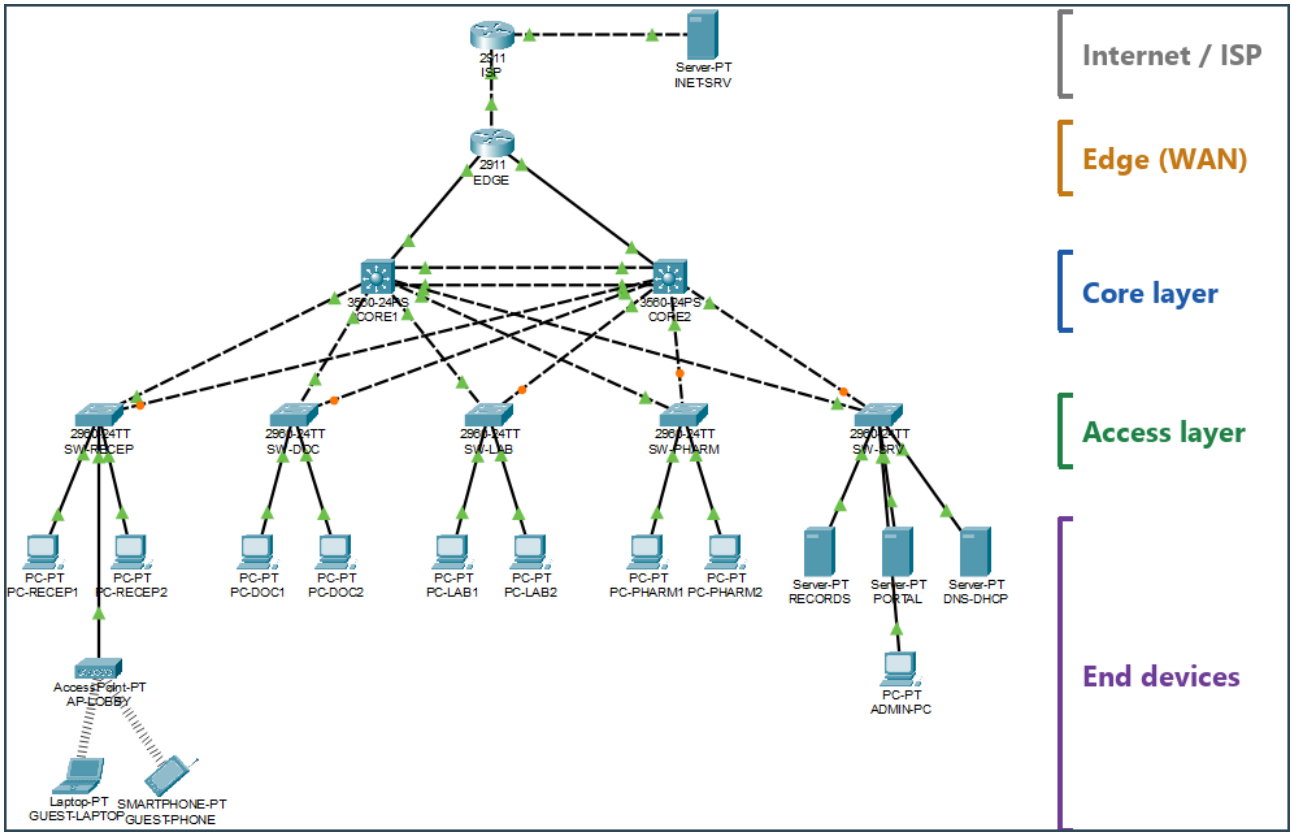


Figure 1: *Completed Packet Tracer topology with the network layers labelled*

5 IP Addressing Table

All internal addresses come from 172.16.10.0/23, split with VLSM, largest subnet first. This avoids wasting addresses: Guest gets a /25 for about 100 users while the Servers VLAN only needs a /28. In every VLAN the first host is the HSRP virtual gateway, the second is CORE1 and the third is CORE2. User PCs and guests get their addresses by DHCP from DNS-DHCP.

Table 2: *IP addressing of the key devices*

Device	Interface	IP address	Subnet mask	Default gateway
PC-DOC1 (example DHCP client)	FastEthernet0	172.16.10.143 (DHCP)	255.255.255.192	172.16.10.129
RECORDS	FastEthernet0	172.16.11.70	255.255.255.240	172.16.11.65
PORTAL	FastEthernet0	172.16.11.71	255.255.255.240	172.16.11.65
DNS-DHCP	FastEthernet0	172.16.11.72	255.255.255.240	172.16.11.65
ADMIN-PC	FastEthernet0	172.16.11.10	255.255.255.224	172.16.11.1
SW-RECEP ... SW-SRV	Vlan99	172.16.11.11 to .15	255.255.255.224	172.16.11.1
CORE1	Vlan10/20/30/40/50/60/99	2nd host of each VLAN	per VLAN	-

Device	Interface	IP address	Subnet mask	Default gateway
CORE2	Vlan10/20/30/40/50/60/99	3rd host of each VLAN	per VLAN	-
CORE1 / CORE2	Fa0/24 (to EDGE)	172.16.11.82 / .86	255.255.255.252	-
EDGE	Gi0/0 / Gi0/1	172.16.11.81 / .85	255.255.255.252	-
EDGE	Gi0/2 (to ISP)	203.0.113.2	255.255.255.252	203.0.113.1 (default route)
ISP	Gi0/0 / Gi0/1	203.0.113.1 / 198.51.100.1	/30 / 255.255.255.0	-
INET-SRV	FastEthernet0	198.51.100.10	255.255.255.0	198.51.100.1

Table 3: *VLAN and VLSM subnet plan*

VLAN	Name	Hosts needed	Subnet	Mask	HSRP gateway (VIP)
60	Guest Wi-Fi	100	172.16.10.0/25	255.255.255.128	172.16.10.1
20	Doctors / OPD	50	172.16.10.128/26	255.255.255.192	172.16.10.129
10	Reception	25	172.16.10.192/27	255.255.255.224	172.16.10.193
30	Laboratory	25	172.16.10.224/27	255.255.255.224	172.16.10.225
99	Management	20	172.16.11.0/27	255.255.255.224	172.16.11.1
40	Pharmacy	12	172.16.11.32/27	255.255.255.224	172.16.11.33
50	Servers	10	172.16.11.64/28	255.255.255.240	172.16.11.65
999	Blackhole (unused ports)	-	-	-	-

6 Configuration and Implementation

The devices were cabled as in Figure 1 and configured from the IOS CLI. The full running configuration of every switch and router is in the project folder (configs/). The main features are summarised below.

6.1 VLANs, access ports and trunks

VLANs 10 to 99 and 999 exist on all seven switches. Department PCs sit on access ports with PortFast and port security, and both uplinks of each access switch are 802.1Q trunks. Unused ports are put in VLAN 999 and shut down.

```
interface FastEthernet0/1                    ! SW-DOC, PC-DOC1
 switchport mode access
 switchport access vlan 20
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 spanning-tree portfast
```

6.2 EtherChannel, Rapid-PVST and HSRP

CORE1 and CORE2 are joined by Port-channel 1 (Gi0/1 and Gi0/2, LACP), carried as a trunk. Rapid-PVST runs on every switch with CORE1 as root bridge for all VLANs, so the blocked uplinks are always the ones to CORE2. Each VLAN has an SVI on both cores and an HSRP group: CORE1 has priority 110 and is Active, CORE2 has 100 and is Standby, with preempt and fast timers (hello 1 s, hold 3 s). The root bridge and the Active gateway are on the same switch, so traffic does not take a detour over the EtherChannel.

```
spanning-tree mode rapid-pvst                ! CORE1
spanning-tree vlan 1,10,20,30,40,50,60,99,999 priority 24576
interface GigabitEthernet0/1                ! same on Gi0/2
 channel-group 1 mode active
interface Vlan20
 ip address 172.16.10.130 255.255.255.192
 ip helper-address 172.16.11.72
 ip access-group DOCTORS-IN in
 standby 20 ip 172.16.10.129
 standby 20 priority 110
 standby 20 preempt
 standby 20 timers 1 3
```

CORE1 | show standby brief

```
CORE1#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Vl110       10   110 P Active  local           172.16.10.195   172.16.10.193
Vl120       20   110 P Active  local           172.16.10.131   172.16.10.129
Vl130       30   110 P Active  local           172.16.10.227   172.16.10.225
Vl140       40   110 P Active  local           172.16.11.35    172.16.11.33
Vl150       50   110 P Active  local           172.16.11.67    172.16.11.65
Vl160       60   110 P Active  local           172.16.10.3     172.16.10.1
Vl199       99   110 P Active  local           172.16.11.3     172.16.11.1
```

CORE2 | show standby brief

```
CORE2#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Vl110       10   100 P Standby 172.16.10.194   local           172.16.10.193
Vl120       20   100 P Standby 172.16.10.130   local           172.16.10.129
Vl130       30   100 P Standby 172.16.10.226   local           172.16.10.225
Vl140       40   100 P Standby 172.16.11.34    local           172.16.11.33
Vl150       50   100 P Standby 172.16.11.66    local           172.16.11.65
Vl160       60   100 P Standby 172.16.10.2     local           172.16.10.1
Vl199       99   100 P Standby 172.16.11.2     local           172.16.11.1
```

Figure 2: *show standby brief: CORE1 Active and CORE2 Standby for all 7 VLAN gateways*

6.3 Routing, NAT and services

OSPF area 0 runs on CORE1 (router ID 1.1.1.1), CORE2 (2.2.2.2) and EDGE. All SVIs are passive, so adjacencies only form on the two routed links to EDGE. EDGE has a static default route to the ISP and advertises it into OSPF with default-information originate. PAT translates the whole 172.16.10.0/23 range to EDGE's public address 203.0.113.2. DNS-DHCP holds five DHCP pools (one per user VLAN, reached through ip helper-address on the cores) and DNS records for records.hospital.local and portal.hospital.local. RECORDS runs HTTP and FTP, PORTAL runs HTTP and HTTPS.

```
router ospf 1                                   ! EDGE
 network 172.16.11.80 0.0.0.3 area 0
 network 172.16.11.84 0.0.0.3 area 0
 default-information originate
ip route 0.0.0.0 0.0.0.0 203.0.113.1
ip nat inside source list NAT-INSIDE interface GigabitEthernet0/2 overload
```

6.4 Security

Named extended ACLs are applied inbound on the user SVIs of both cores, so the policy still holds after a failover (Table 4). Every user ACL also permits DHCP, DNS to 172.16.11.72 and ICMP echo-reply, then denies the rest of 172.16.10.0/23 and permits the internet. Switches and EDGE accept only SSH version 2 on the VTY lines, only from VLAN 99 (access-class MGMT-SSH), with a local user, enable secret and service password-encryption.

Table 4: *Department access policy (ACLs inbound on the core SVIs)*

Source VLAN (ACL)	Allowed inside the hospital	Blocked
Doctors 20 (DOCTORS-IN)	Everything: RECORDS, PORTAL, other departments	Management VLAN 99
Reception 10 (RECEPTION-IN)	PORTAL web (HTTP, HTTPS)	RECORDS, other departments
Laboratory 30 (LAB-IN)	PORTAL web, RECORDS FTP (upload results)	RECORDS web, other departments
Pharmacy 40 (PHARMACY-IN)	PORTAL web, RECORDS HTTP (view prescriptions)	RECORDS FTP and HTTPS, other departments
Guest 60 (GUEST-IN)	Only DHCP and DNS	All of 172.16.10.0/23
Management 99 (no ACL)	Everything, SSH to all devices	-

7 Testing and Results

Tests were run in Realtime mode with ping, nslookup, the PC web browser, FTP and SSH clients. The expected outcome of each test comes from the design: allowed traffic must work, blocked traffic must fail, and a failure in the core must cost only a few packets. Table 5 summarises the results.

Table 5: *Test summary*

Source	Destination	Test performed	Result
One host per VLAN (6)	RECORDS, PORTAL, DNS-DHCP, INET-SRV, PC-RECEP2, PC-DOC2, ADMIN-PC	Ping matrix (42 cells)	Success: all 42 cells matched the ACL policy

Source	Destination	Test performed	Result
PC-RECEP1, PC-DOC1, PC-LAB1, PC-PHARM1, GUEST-LAPTOP	PORTAL, RECORDS, INET-SRV	Web browser (12 HTTP/HTTPS checks)	Success: 12/12 as expected (e.g. Reception gets PORTAL, RECORDS times out)
PC-LAB1 / PC-PHARM1	RECORDS	FTP login	Success: Lab logged in; Pharmacy blocked as intended
DHCP clients	DNS-DHCP	DHCP lease, nslookup	Success: correct address, gateway and DNS on every client
PC-DOC1	CORE2 Vlan20 (via Po1)	60 pings, one EtherChannel member shut	Success: 60/60 replies, Po1 stayed up on Gi0/2
PC-DOC1	RECORDS	200 pings, CORE1 powered off	Success: 198/200 replies, CORE2 Active on all 7 groups
GUEST-LAPTOP	RECORDS, PC-DOC1, INET-SRV	Ping, web, NAT check	Success: internal blocked (8 ACL denies), internet 20/20 via PAT
ADMIN-PC / PC-DOC1	SW-RECEP 172.16.11.11	SSH and Telnet	Success: SSH from VLAN 99 only, Telnet refused

7.1 EtherChannel member failure

PC-DOC1 pinged CORE2's VLAN 20 address. Every packet had to cross Po1 because SW-DOC's uplink to CORE2 is blocked. While the ping ran, Gi0/1 on CORE1 was shut down. Po1 stayed up (SU) on Gi0/2 and the recorded test lost no packets (60/60). In the screenshot run, 1 of 60 was lost when Gi0/1 came back up and HSRP on CORE2 briefly flapped for two VLANs.

PC-DOC1 | ping -n 60 172.16.10.131

```
C:\>ping -n 60 172.16.10.131

Pinging 172.16.10.131 with 32 bytes of data:

Reply from 172.16.10.131: bytes=32 time<1ms TTL=255
Reply from 172.16.10.131: bytes=32 time<1ms TTL=255
Reply from 172.16.10.131: bytes=32 time<1ms TTL=255
Reply from 172.16.10.131: bytes=32 time=7ms TTL=255
Reply from 172.16.10.131: bytes=32 time<1ms TTL=255
Reply from 172.16.10.131: bytes=32 time<1ms TTL=255
Reply from 172.16.10.131: bytes=32 time<1ms TTL=255
Reply from 172.16.10.131: bytes=32 time<1ms TTL=255
Reply from 172.16.10.131: bytes=32 time<1ms TTL=255
Reply from 172.16.10.131: bytes=32 time<1ms TTL=255
Reply from 172.16.10.131: bytes=32 time<1ms TTL=255
Reply from 172.16.10.131: bytes=32 time<1ms TTL=255
Reply from 172.16.10.131: bytes=32 time<1ms TTL=255
Reply from 172.16.10.131: bytes=32 time<1ms TTL=255
Reply from 172.16.10.131: bytes=32 time<1ms TTL=255
Reply from 172.16.10.131: bytes=32 time<1ms TTL=255
Reply from 172.16.10.131: bytes=32 time=1ms TTL=255
Reply from 172.16.10.131: bytes=32 time<1ms TTL=255
Reply from 172.16.10.131: bytes=32 time<1ms TTL=255
Reply from 172.16.10.131: bytes=32 time<1ms TTL=255
Reply from 172.16.10.131: bytes=32 time<1ms TTL=255
```

CORE1 | shutdown Gi0/1

```
CORE1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
CORE1(config)#int g0/1
CORE1(config-if)#shutdown

CORE1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
```

CORE1 | show etherchannel summary

```
Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           LACP   Gig0/1(D)  Gig0/2(P)
```

Figure 3: *Ping keeps replying while Gi0/1 is shut; Po1 stays up with Gig0/1(D) Gig0/2(P)*

7.2 Core switch failure (HSRP failover)

PC-DOC1 sent 200 pings to RECORDS and CORE1 was powered off mid-way. Both the VLAN 20 and the VLAN 50 gateway had to move to CORE2. Only 2 packets were lost (198/200, about 2 seconds), which matches the 3-second HSRP hold time. With default timers (hold 10 s) about 10 packets would be lost. The screenshot run lost 3 packets. When CORE1 came back it took the Active role back (preempt) with 3 of 150 packets lost. The PCs kept the same gateway address the whole time.

PC-DOC1 | ping during CORE1 power-off

```
C:\>ping -n 200 172.16.11.70

Pinging 172.16.11.70 with 32 bytes of data:

Reply from 172.16.11.70: bytes=32 time<1ms TTL=127
Reply from 172.16.11.70: bytes=32 time<1ms TTL=127
Reply from 172.16.11.70: bytes=32 time<1ms TTL=127
Reply from 172.16.11.70: bytes=32 time<1ms TTL=127
Reply from 172.16.11.70: bytes=32 time<1ms TTL=127
Request timed out.
Request timed out.
Request timed out.
Reply from 172.16.11.70: bytes=32 time<1ms TTL=127
Reply from 172.16.11.70: bytes=32 time<1ms TTL=127
Reply from 172.16.11.70: bytes=32 time<1ms TTL=127
Reply from 172.16.11.70: bytes=32 time<1ms TTL=127
Reply from 172.16.11.70: bytes=32 time<1ms TTL=127
```

CORE2 | show standby brief (CORE1 off)

```
CORE2#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   100 P Active   local           unknown         172.16.10.193
Vl20        20   100 P Active   local           unknown         172.16.10.129
Vl30        30   100 P Active   local           unknown         172.16.10.225
Vl40        40   100 P Active   local           unknown         172.16.11.33
Vl50        50   100 P Active   local           unknown         172.16.11.65
Vl60        60   100 P Active   local           unknown         172.16.10.1
Vl99        99   100 P Active   local           unknown         172.16.11.1
CORE2#
```

Figure 4: *Left: ping loses 3 packets during the failover. Right: CORE2 is now Active for all 7 groups*

7.3 Guest isolation and internet access

From the guest laptop on the lobby Wi-Fi, pings and web requests to RECORDS and to a Doctors PC failed with "Destination host unreachable" from the guest VLAN's own gateway, which shows that the core ACL dropped them (8 deny matches on GUEST-IN). The internet web server loaded and 20/20 pings succeeded, translated by PAT on EDGE.

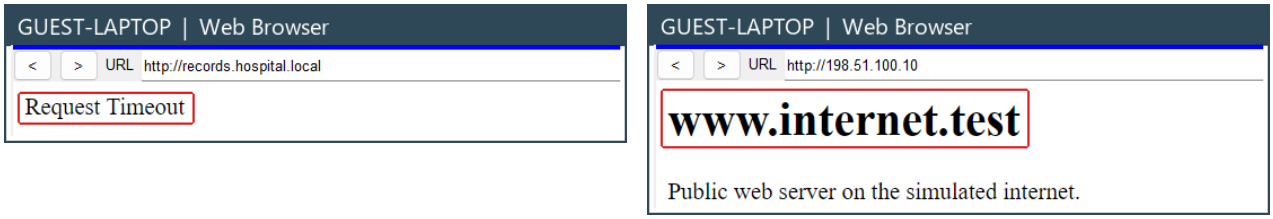


Figure 5: *Guest laptop: the records server times out (left), the internet server loads (right)*

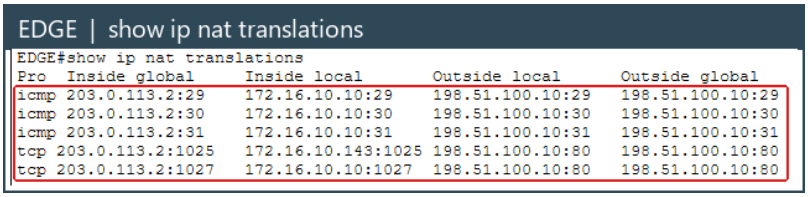


Figure 6: *PAT on EDGE: inside local 172.16.10.10 leaves as inside global 203.0.113.2*

8 Conclusion

All six objectives were achieved and each one was confirmed by a test in Section 7. The departments are separated into VLANs with a VLSM plan, every access policy in Table 4 behaved as designed, losing an EtherChannel link cost no traffic, and losing a whole core switch cost about 2 seconds without any change on the PCs. Staff and guests reach the internet through PAT, while guests cannot reach any hospital system. The main lesson was that the features must agree with each other: the spanning tree root has to be the same switch as the HSRP Active gateway, and the order of ACL lines decides what is really allowed.

Limitations and possible improvements:

- EDGE and the single ISP link are a single point of failure for internet access. A second edge router and ISP would remove it.
- Packet Tracer does not save the HSRP timers (standby timers 1 3), so they must be typed again after a reload. Real switches keep them.
- CORE1 is Active for every VLAN, so CORE2 only forwards after a failure. HSRP load sharing would use both cores.
- After a core power cycle, spanning tree once stayed in a wrong state until the CORE2 downlinks were reset. During that state a ping showed duplicate replies (a short loop).
- The guest Wi-Fi is an open network; a real deployment would use WPA2/WPA3 with a captive portal. Other future work: AAA with RADIUS, syslog and SNMP monitoring, IPv6 and a firewall at the edge.

References

- Cisco Networking Academy, Cisco Packet Tracer 9.0.1 documentation and help files.
- Cisco Systems, Catalyst 3560 Switch Software Configuration Guide (VLANs, EtherChannel, Rapid-PVST, HSRP, ACLs).
- Cisco Systems, Cisco 2900 Series ISR Software Configuration Guide (OSPF, NAT/PAT, SSH).
- Course textbook and lab manuals, Computer Networks (PE-7), Manipal Institute of Technology, Bengaluru.

Appendix A: Additional Screenshots

Further evidence from the simulation. Each screenshot is cropped to the relevant output and the key lines are boxed in red. Screenshots already shown in the report are not repeated.

SW-DOC | show vlan brief

```
SW-DOC#show vlan brief

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active
10   RECEPTION                         active
20   DOCTORS                           active    Fa0/1, Fa0/2
30   LAB                               active
40   PHARMACY                          active
50   SERVERS                           active
60   GUEST                             active
99   MANAGEMENT                        active
999  BLACKHOLE                         active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                 Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                 Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                 Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                 Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                 Fa0/23, Fa0/24
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
SW-DOC#
```

Figure A1: *SW-DOC show vlan brief: Fa0/1-2 in VLAN 20, unused ports in VLAN 999*

SW-DOC | show interfaces trunk

```
SW-DOC#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1
Gig0/2      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      1-1005
Gig0/2      1-1005

Port        Vlans allowed and active in management domain
Gig0/1      1,10,20,30,40,50,60,99,999
Gig0/2      1,10,20,30,40,50,60,99,999

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      1,10,20,30,40,50,60,99,999
Gig0/2      none

SW-DOC#
```

Figure A2: *SW-DOC show interfaces trunk: Gi0/2 forwards no VLANs (blocked by spanning tree)*

CORE1 | show spanning-tree vlan 20

```
CORE1#show spanning-tree vlan 20
VLAN0020
  Spanning tree enabled protocol rstp
  Root ID     Priority    24596
              Address     00D0.D36E.6996
              This bridge is the root
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    24596  (priority 24576 sys-id-ext 20)
              Address     00D0.D36E.6996
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/5            Desg FWD 19        128.5    P2p
Fa0/3            Desg FWD 19        128.3    P2p
Fa0/4            Desg FWD 19        128.4    P2p
Fa0/2            Desg FWD 19        128.2    P2p
Po1              Desg FWD 3         128.27   P2p
```

Figure A3: *CORE1 show spanning-tree vlan 20: this bridge is the root*

CORE1 | show etherchannel summary

```
Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           LACP   Gig0/1(P) Gig0/2(P)
```

Figure A4: *CORE1 show etherchannel summary: Po1(SU) with both members bundled (P)*

EDGE | show ip ospf neighbor

```
EDGE#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/BDR        00:00:39    172.16.11.86    GigabitEthernet0/1
1.1.1.1           1   FULL/BDR        00:00:39    172.16.11.82    GigabitEthernet0/0
```

Figure A5: *EDGE show ip ospf neighbor: two FULL neighbours, 1.1.1.1 and 2.2.2.2*

PC-DOC1 | Desktop > IP Configuration (DHCP)

DHCP	Static
IPv4 Address	172.16.10.143
Subnet Mask	255.255.255.192
Default Gateway	172.16.10.129
DNS Server	172.16.11.72

Figure A6: *PC-DOC1 lease from DNS-DHCP: address, mask, gateway 172.16.10.129 and DNS 172.16.11.72*

CORE1 | show access-lists (part)

```
CORE1#show access-lists
Extended IP access list GUEST-IN
    10 permit udp any any eq bootps (10 match(es))
    20 permit udp any host 172.16.11.72 eq domain
    30 permit icmp any any echo-reply
    40 deny ip any 172.16.10.0 0.0.1.255 (24 match(es))
    50 permit ip any any (1246 match(es))
Extended IP access list DOCTORS-IN
    10 deny ip 172.16.10.128 0.0.0.63 172.16.11.0 0.0.0.31 (7 match(es))
    20 permit ip any any (1284 match(es))
```

Figure A7: *CORE1 show access-lists: match counts on the deny lines after blocked pings*

PC-RECEP1 | Web Browser

< > URL http://portal.hospital.local

PORTAL - Hospital Staff Portal

Rosters, notices and appointments for all departments.

Figure A8: *PC-RECEP1 opens the staff portal (allowed)*

PC-RECEP1 | Web Browser

< > URL http://records.hospital.local

Request Timeout

Figure A9: *PC-RECEP1 cannot open the records server (blocked by RECEPTION-IN)*

PC-LAB1 | Command Prompt

```
C:\>ftp 172.16.11.70
Trying to connect...172.16.11.70
Connected to 172.16.11.70
220- Welcome to PT Ftp server
Username:labftp
331- Username ok, need password
Password:******
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 172.16.11.70:
0    : asa842-k8.bin                                5571584
1    : asa923-k8.bin                                30468096
2    : c1841-advipservicesk9-mz.124-15.T1.bin       33591768
3    : c1841-ipbase-mz.123-14.T7.bin                13832032
4    : c1841-ipbasek9-mz.124-12.bin                 16599160
```

Figure A10: *PC-LAB1 logs in to the RECORDS FTP server (230 Logged in)*

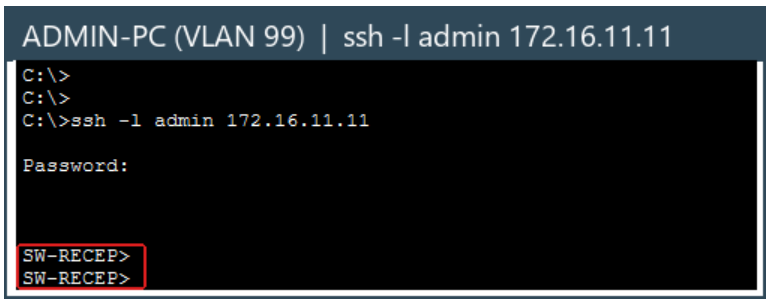


Figure A11: *ADMIN-PC in VLAN 99 reaches SW-RECEP over SSH*

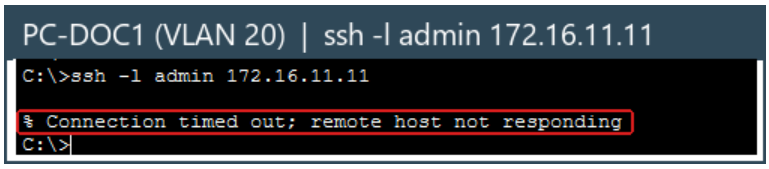


Figure A12: *PC-DOC1 in VLAN 20 cannot open SSH to the switch (connection timed out)*